A07 AI in Blockchain Module Reflections

1. Industry Overview

Before studying this module, I mostly associated blockchain with Bitcoin and cryptocurrency. However, after reviewing the lecture material, I learned that blockchain is much larger than digital currency. Blockchain is a decentralized digital ledger that allows information and transactions to be stored securely across many computers instead of being controlled by a single organization. The blockchain industry serves many different areas, including finance, healthcare, supply chain management, education, and government services. Organizations use blockchain to improve security, increase transparency, and reduce the possibility of fraud or unauthorized changes to data. The module discussed examples such as patient records, food tracking systems, smart contracts, and credential verification.

Today, this industry is becoming increasingly important because businesses depend heavily on digital information. Data breaches, cyberattacks, and concerns about trust continue to grow. Blockchain offers a way to protect information while allowing multiple parties to share data securely.

What I found especially interesting is how blockchain and artificial intelligence can work together. AI systems require large amounts of reliable data, while blockchain helps ensure that the data is accurate and secure. This relationship makes the industry very relevant as AI becomes more common in many fields.

2. Current AI Trends

One trend that stood out to me is the use of blockchain to improve AI data integrity. The lecture explained that AI systems depend on trustworthy data, and if the data is manipulated or incorrect, the AI may produce poor decisions. Blockchain creates permanent records that make it easier to verify where data comes from and whether it has been altered.

I found this concept interesting because it addresses one of the major concerns about AI. Many people trust AI systems without knowing where their information comes from. Blockchain can create an audit trail that allows organizations to verify data sources and understand how AI decisions are made.

Another important trend is the use of AI and blockchain in supply chain management. The Walmart example from the module helped me understand how these technologies work in real businesses. Blockchain tracks food products throughout the supply chain, while AI analyzes customer demand and inventory levels. This helps reduce waste, improve efficiency, and quickly identify contaminated products.

The IPwe Global Patent Registry was another example that surprised me. I had never considered how AI and blockchain could be used for patent management. Blockchain secures ownership records while AI analyzes market trends and patent values. This shows that these technologies can solve problems outside of finance and cryptocurrency.

Finally, decentralized AI and federated learning appear to be important for future developments. Instead of storing all information in one place, organizations can collaborate while keeping sensitive data private. This seems especially useful in industries such as healthcare and banking.

3. Ethical Implications

One ethical issue that concerns me is privacy. Blockchain is designed to keep records permanently, which creates a conflict when personal information is involved. While transparency can increase trust, people may not want their information stored forever. This creates challenges when trying to balance privacy with security. The module discussed solutions such as encryption and permissioned blockchains, but I think this issue will continue to be important as more organizations adopt these technologies. Another concern is bias in artificial intelligence. AI systems learn from existing data, and if the data contains bias, the AI may produce unfair decisions. Blockchain can help track where data comes from, but it cannot automatically remove bias from the information itself.

I also think transparency is an important issue. AI systems often operate like a "black box," meaning users do not fully understand how decisions are made. Blockchain can create records of decisions and make systems more accountable, but the technology can still be difficult for average users to understand. Finally, automation may affect certain jobs. Smart contracts and AI systems can perform tasks that previously required human workers. Although new jobs may be created, workers may need additional training to adapt to these changes. After studying this topic, I believe that ethical considerations are just as important as technology itself. Building advanced systems is valuable, but ensuring fairness, privacy, and accountability is equally important.

4. Future Trends

Looking ahead, I think AI and blockchain will become much more connected during the next five to ten years. The module introduced concepts such as decentralized AI marketplaces, federated learning, and edge computing, which all seem promising.

One trend that interested me was federated learning. Instead of sending sensitive information to centralized servers, organizations can train AI systems locally while sharing only model updates. This could help protect privacy while still allowing organizations to benefit from AI. I also think blockchain may become increasingly important for AI transparency. Governments and organizations are beginning to create regulations for AI systems, and blockchain may help provide the documentation and accountability that regulators require.

However, there are still several risks. Scalability remains a challenge because some blockchain networks process transactions slowly. Energy consumption is another concern, especially with older systems such as Proof of Work. Although newer methods like Proof of Stake are more efficient, these problems still need to be addressed.

Another challenge is that many blockchain systems are difficult for average users to understand. If these technologies become more common, education and user-friendly designs will become increasingly important.

Overall, I believe the future offers many opportunities. AI and blockchain together could improve healthcare, finance, supply chains, and many other industries, but developers must continue addressing the technical and ethical challenges.

5. Personal Reflection

This module changed my understanding of blockchain more than I expected. Before taking this class, I mainly thought about Bitcoin whenever I heard the word blockchain. I did not realize that technology could be used for healthcare records, food safety, patents, or artificial intelligence. The Walmart case study was one of the most interesting examples because it showed how these technologies are already being used in real businesses. I was also surprised to learn that blockchain can help make AI systems more transparent by creating audit trails and protecting data integrity.

Another thing that stood out to me was the relationship between AI and trust. AI systems often make decisions that people do not fully understand, and blockchain may help increase confidence in those decisions by creating permanent records. I would consider working in this industry because it combines several areas that interest me, including artificial intelligence, cybersecurity, and data analysis. The field appears to have many opportunities for growth, and the problems being solved are meaningful and practical.

At the same time, this module showed me that technology is not only about creating new systems. Developers must also think about privacy, fairness, and ethical responsibilities. I believe these lessons will be valuable not only in future classes but also in my future career.

Overall, this module helped me understand that blockchain is much more than cryptocurrency. It is a technology that can improve trust, security, and transparency, especially when combined with artificial intelligence.